Honeypot Investigation Report

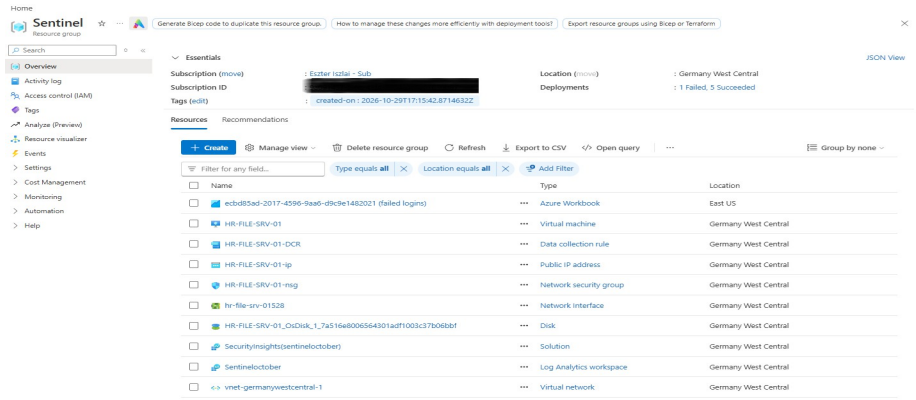
Sentineloctober

November 01, 2026

By Eszter Iszlai

Executive Summary

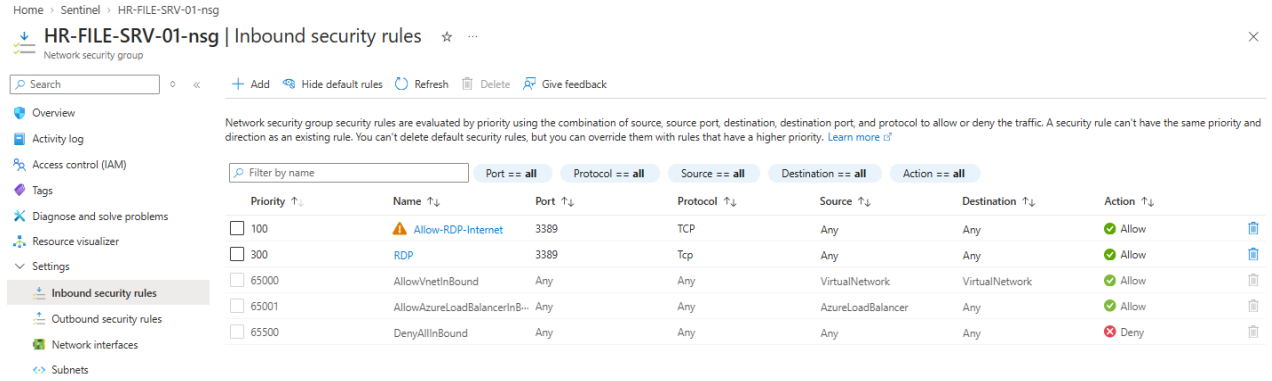
The objective of this project was to deploy and monitor a Windows VM acting as a honeypot using Microsoft Sentinel.
The VM was connected to a Log Analytics Workspace and configured to collect Windows Security Events.
During the project, login attempts were monitored, a peer-to-peer attack simulation was conducted, and additional telemetry was added through Sysmon to improve forensic visibility.

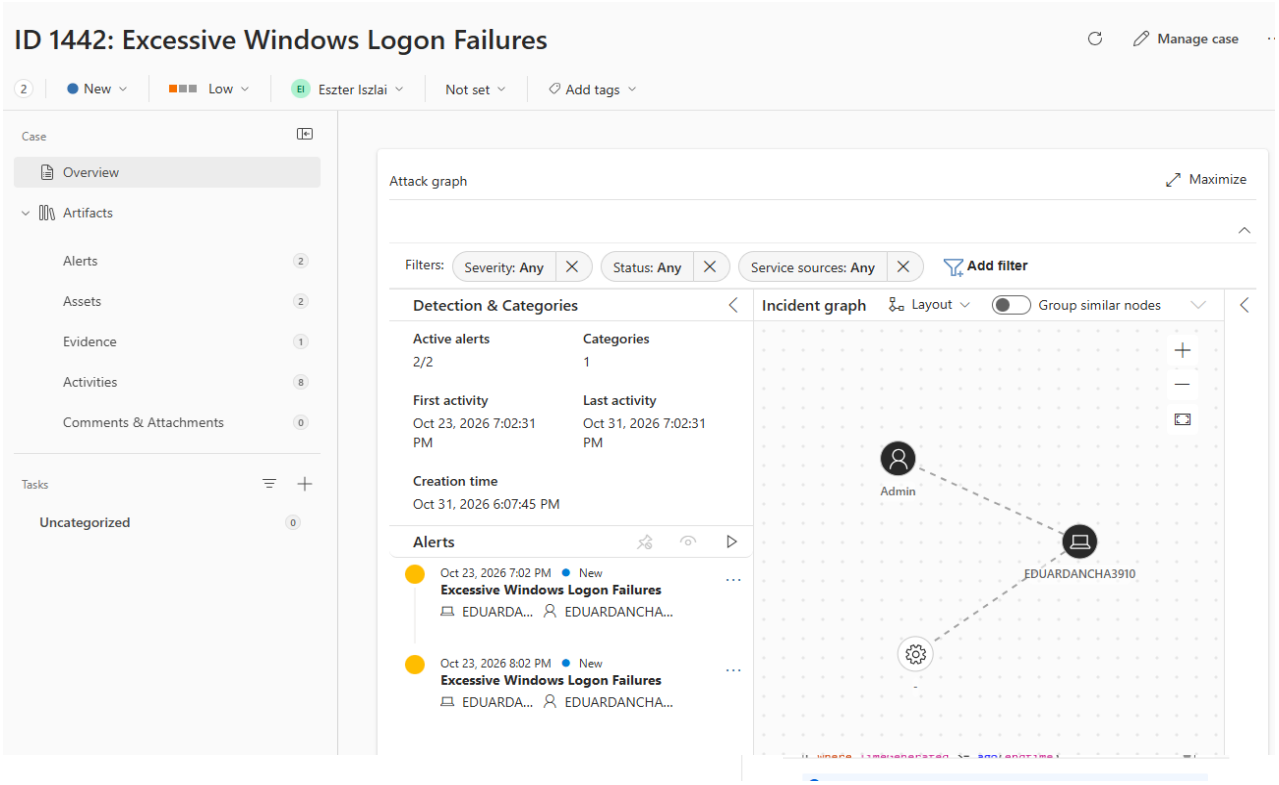


The setup of the Honeypot

<table>
<tr><th>Task</th><th>Notes</th></tr>
<tr><td>Create Windows VM</td><td>Created HR-FILE-SRV-01</td></tr>
<tr><td>Create Public IP</td><td>VM has a public IP address assigned</td></tr>
<tr><td>Create Network Security Group</td><td>HR-FILE-SRV-01-nsg created</td></tr>
<tr><td>Open RDP to Internet</td><td>Port 3389/TCP allowed from Any source</td></tr>
<tr><td>Use strong administrator password</td><td>Password is intentionally strong to prevent compromise</td></tr>
<tr><td>Create Log Analytics Workspace</td><td>Workspace Sentineloctober created</td></tr>
<tr><td>Enable Microsoft Sentinel</td><td>Sentinel workspace is accessible and connector configuration has begun</td></tr>
<tr><td>Install Windows Security Events via AMA solution</td><td>Connector opened and being configured</td></tr>
<tr><td>Install Azure Monitor Agent (AMA)</td><td>Screenshot shows AzureMonitorWindowsAgent with Provisioning succeeded</td></tr>
<tr><td>Create Data Collection Rule (DCR)</td><td>DCR creation wizard completed</td></tr>
<tr><td>Assign VM to DCR</td><td>VM associated with the rule</td></tr>
<tr><td>Collect All Security Events</td><td>Select All Security Events during DCR configuration</td></tr>
<tr><td>Verify events arriving in Sentinel</td><td>Check SecurityEvent logs after configuration</td></tr>
<tr><td>Observe RDP brute-force activity</td><td>Should begin after logging is configured and bots discover the VM</td></tr>
<tr><td>Smoke Test</td><td>Failed login detection verified</td></tr>
<tr><td>Active Monitoring</td><td>Azure Monitor Agent functioning, Data Collection Rule functioning, Security Event collection working, Sentinel analytics rule enabled</td></tr>
<tr><td>Attacker Detected</td><td>
<table>
<tr><th>Item</th><th>Finding</th></tr>
<tr><td>Incident Name</td><td>Excessive Windows Logon Failures</td></tr>
<tr><td>Incident ID</td><td>1442</td></tr>
<tr><td>Severity</td><td>Low</td></tr>
<tr><td>Status</td><td>Active</td></tr>
<tr><td>Owner</td><td>Eszter Iszlai</td></tr>
<tr><td>Target Host</td><td>EDUARDANCHA3910</td></tr>
<tr><td>Target Account</td><td>Admin / Administrator</td></tr>
<tr><td>Source IP</td><td>95.84.208.84</td></tr>
<tr><td>Event ID</td><td>4625 (Failed Logon)</td></tr>
<tr><td>Logon Type</td><td>3 - Network</td></tr>
<tr><td>Detection Source</td><td>Microsoft Sentinel Analytics Rule</td></tr>
</table>
</td></tr>
<tr><td>Incident Generated</td><td>Microsoft Sentinel successfully detected and generated an incident for repeated failed Windows logon attempts. Investigation of Event ID 4625 logs identified multiple authentication failures originating from IP address 95.84.208.84 targeting the local administrative account on host EDUARDANCHA3910. The incident was assigned,</td></tr>
</table>

<table>
<tr><td></td><td>reviewed, and marked Active, demonstrating the full incident triage and investigation workflow in Microsoft Sentinel.</td></tr>
<tr><td>Attack Timeline</td><td><table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>VM exposed to internet via RDP</td></tr>
<tr><td>2</td><td>External actor attempted logins</td></tr>
<tr><td>3</td><td>Windows generated Event ID 4625 logs</td></tr>
<tr><td>4</td><td>Logs forwarded to Log Analytics</td></tr>
<tr><td>5</td><td>Sentinel Analytics Rule triggered</td></tr>
<tr><td>6</td><td>Sentinel created incident</td></tr>
<tr><td>7</td><td>Incident assigned to analyst</td></tr>
<tr><td>8</td><td>Incident status changed to Active</td></tr>
<tr><td>9</td><td>Source IP investigated</td></tr>
<tr><td>10</td><td>Brute-force activity confirmed</td></tr>
</table></td></tr>
</table>


Home › Sentinel › HR-FILE-SRV-01-nsg
HR-FILE-SRV-01-nsg | Inbound security rules
Network security group
Search
Add
Hide default rules
Refresh
Delete
Give feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. Learn more
Filter by name
Port == all
Protocol == all
Source == all
Destination == all
Action == all
Priority
Name
Port
Protocol
Source
Destination
Action
100 Allow-RDP-Internet 3389 TCP Any Any Allow
300 RDP 3389 Tcp Any Any Allow
65000 AllowVnetInBound Any Any VirtualNetwork VirtualNetwork Allow
65001 AllowAzureLoadBalancerInB... Any Any AzureLoadBalancer Any Allow
65500 DenyAllInBound Any Any Any Any Deny


ID 1442: Excessive Windows Logon Failures
Manage case
New
Low
Eszter Iszlai
Not set
Add tags
Case
Overview
Artifacts
Alerts 2
Assets 2
Evidence 1
Activities 8
Comments & Attachments 0
Tasks
Uncategorized 0
Attack graph
Maximize
Filters:
Severity: Any
Status: Any
Service sources: Any
Add filter
Detection & Categories
Active alerts
2/2
Categories
1
First activity
Oct 23, 2026 7:02:31 PM
Last activity
Oct 31, 2026 7:02:31 PM
Creation time
Oct 31, 2026 6:07:45 PM
Alerts
Oct 23, 2026 7:02 PM New
Excessive Windows Logon Failures
EDUARDA... EDUARDANCHA...
Oct 23, 2026 8:02 PM New
Excessive Windows Logon Failures
EDUARDA... EDUARDANCHA...
Incident graph
Layout
Group similar nodes
Admin
EDUARDANCHA3910

Statistics

Total Activity Observed

The honeypot recorded:

- Successful logons (Event ID 4624) - 257
- Failed logons (Event ID 4625) - 484
- Account creation events (Event ID 4720) - 1
- Administrator group membership changes (Event ID 4732) - 2
- Sysmon Process Creation events (Event ID 1) - 49

Frequently used queries

Successful Logons

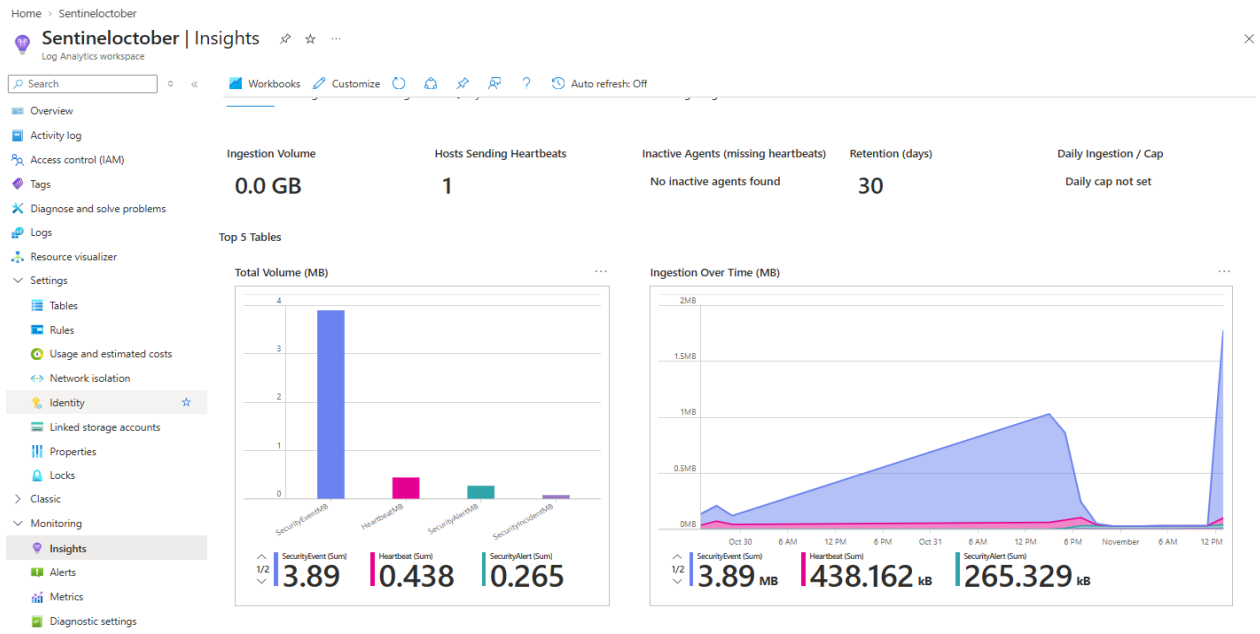
```
SecurityEvent
| where EventID == 4624
| sort by TimeGenerated desc
```

Failed Logons

```
SecurityEvent
| where EventID == 4625
| sort by TimeGenerated desc
```

Top Targeted Usernames

```
SecurityEvent
| where EventID == 4625
| summarize Attempts=count() by Account
| top 10 by Attempts
```

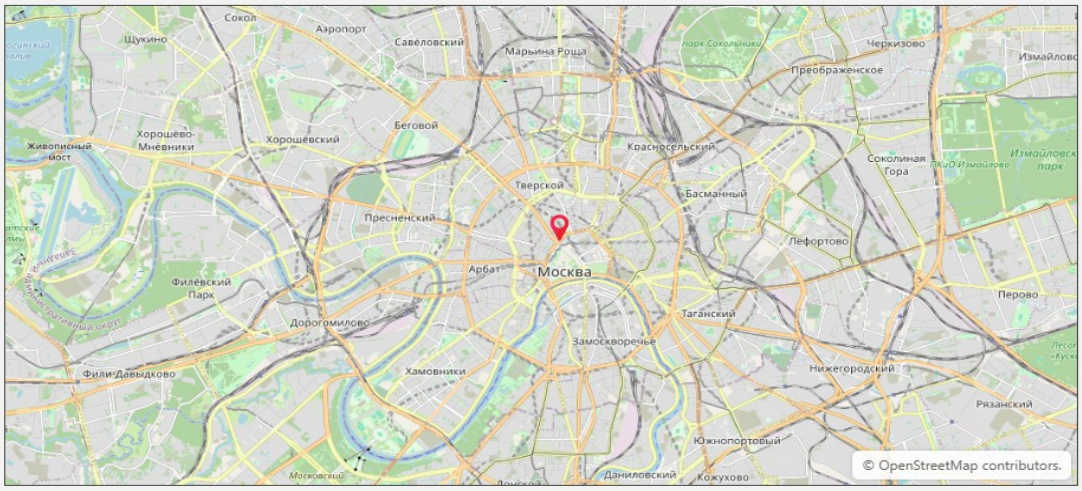


Most Active Country - the attacker is from Moscow, Russia

GEOLOCATION

Home IP Geolocation API Fre

IP Information - 95.84.208.84



W3C Geolocation API Demo

Country	Region	City
Russian Federation	Moskva	Moscow
ZIP or Postal Code	**Latitude**	**Longitude**
101990	55.752258	37.615471
ISP	**Domain Name**	**Usage Type**
National Cable Networks	nationalcablenetworks.ru [WHOIS] [Check Mail Server]	Fixed Line ISP
Weather		

The Classmate Attack

Scenario
A classmate was provided with valid RDP credentials and successfully logged into the VM.
The following commands were executed:
systeminfo
whoami /priv
whoami /groups
arp -a
net user xxxxx PasswQrD2026 /add
net localgroup administrators xxxxx /add

Timeline

Time	Evidence	Interpretation
11:34	Event 4624, LogonType 10	Attacker logged in via RDP
11:35	systeminfo	Host discovery
11:35	whoami /priv	Privilege enumeration
11:35	whoami /groups	Group enumeration
11:35	arp -a	Network discovery
11:36	Event 4720	Created user **xxxxx**
11:36	Event 4732	Added **xxxxx** to Administrators

Investigation Queries

Purpose	Event ID	KQL Query
Find all successful logins	4624	SecurityEvent | where EventID == 4624 | sort by TimeGenerated desc
Find RDP logins only	4624 (LogonType 10)	SecurityEvent | where EventID == 4624 | where LogonType == 10 | project TimeGenerated, Account, Computer, IpAddress | sort by TimeGenerated desc
Find failed logins	4625	SecurityEvent | where EventID == 4625 | project TimeGenerated, Account, Computer, IpAddress | sort by TimeGenerated desc
Find user account creation	4720	SecurityEvent | where EventID == 4720 | sort by TimeGenerated desc
Search specifically for user "xxxxx"	4720	SecurityEvent | where EventID == 4720 | search "xxxxx"
Find members added to local groups	4732	SecurityEvent | where EventID == 4732 | sort by TimeGenerated desc
Search for "xxxxx" being added to a group	4732	SecurityEvent | where EventID == 4732 | search "xxxxx"
Count logins by IP address	4624	SecurityEvent | where EventID == 4624 | summarize count() by IpAddress | sort by count_ desc

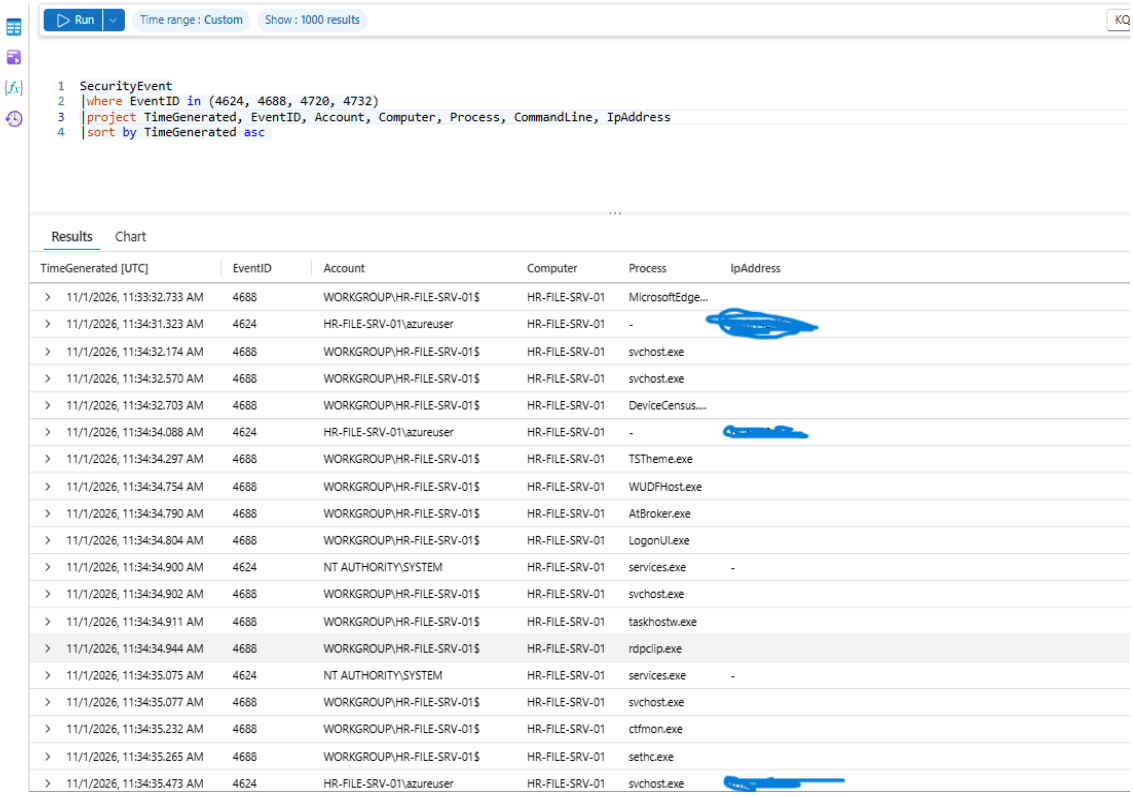
Purpose	Event ID	KQL Query
Count logins by account	4624	SecurityEvent | where EventID == 4624 | summarize count() by Account | sort by count_ desc
Check whether Process Creation auditing is enabled	4688	SecurityEvent | where EventID == 4688 | sort by TimeGenerated desc
Search for systeminfo execution	4688	SecurityEvent | where EventID == 4688 | search "systeminfo"
Search for whoami execution	4688	SecurityEvent | where EventID == 4688 | search "whoami"
Search for net user commands	4688	SecurityEvent | where EventID == 4688 | search "net"
If Sysmon exists, view process creation *this was created after the event*	Sysmon Event 1	Event | where EventLog == "Microsoft-Windows-Sysmon/Operational" | where EventID == 1

Reconstructed events

Step	Action	Evidence
1	RDP Login	Event ID 4624
2	Host Enumeration	systeminfo
3	Privilege Enumeration	whoami /priv
4	Group Enumeration	whoami /groups
5	Network Discovery	arp -a
6	Create User	Event ID 4720
7	Add User to Administrators	Event ID 4732

Attack Classification

Command	ATT&CK-style Purpose
systeminfo	Host Discovery
whoami /priv	Privilege Discovery
whoami /groups	Account Discovery
arp -a	Network Discovery
net user xxxxx ... /add	Persistence
net localgroup administrators xxxxx /add	Privilege Escalation / Persistence



Run | Time range : Custom | Show : 1000 results | KQ

```
1  SecurityEvent
2  |where EventID in (4624, 4688, 4720, 4732)
3  |project TimeGenerated, EventID, Account, Computer, Process, CommandLine, IpAddress
4  |sort by TimeGenerated asc
```

Results | Chart

TimeGenerated [UTC]	EventID	Account	Computer	Process	IpAddress
11/1/2026, 11:33:32.733 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	MicrosoftEdge...	
11/1/2026, 11:34:31.323 AM	4624	HR-FILE-SRV-01\azureuser	HR-FILE-SRV-01	-	[redacted]
11/1/2026, 11:34:32.174 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	svchost.exe	
11/1/2026, 11:34:32.570 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	svchost.exe	
11/1/2026, 11:34:32.703 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	DeviceCensus....	
11/1/2026, 11:34:34.088 AM	4624	HR-FILE-SRV-01\azureuser	HR-FILE-SRV-01	-	[redacted]
11/1/2026, 11:34:34.297 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	TSTheme.exe	
11/1/2026, 11:34:34.754 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	WUDFHost.exe	
11/1/2026, 11:34:34.790 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	AtBroker.exe	
11/1/2026, 11:34:34.804 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	LogonUI.exe	
11/1/2026, 11:34:34.900 AM	4624	NT AUTHORITY\SYSTEM	HR-FILE-SRV-01	services.exe	-
11/1/2026, 11:34:34.902 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	svchost.exe	
11/1/2026, 11:34:34.911 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	taskhostw.exe	
11/1/2026, 11:34:34.944 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	rdpclip.exe	
11/1/2026, 11:34:35.075 AM	4624	NT AUTHORITY\SYSTEM	HR-FILE-SRV-01	services.exe	-
11/1/2026, 11:34:35.077 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	svchost.exe	
11/1/2026, 11:34:35.232 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	ctfmon.exe	
11/1/2026, 11:34:35.265 AM	4688	WORKGROUP\HR-FILE-SRV-01$	HR-FILE-SRV-01	sethc.exe	
11/1/2026, 11:34:35.473 AM	4624	HR-FILE-SRV-01\azureuser	HR-FILE-SRV-01	svchost.exe	[redacted]

```
NT AUTHORITY\This Organization                  Well-known group S-1-5-15     Mandatory group, Enabled
 by default, Enabled group
NT AUTHORITY\Local account                      Well-known group S-1-5-113    Mandatory group, Enabled
 by default, Enabled group
LOCAL                                           Well-known group S-1-2-0      Mandatory group, Enabled
 by default, Enabled group
NT AUTHORITY\NTLM Authentication                Well-known group S-1-5-64-10  Mandatory group, Enabled
 by default, Enabled group
Mandatory Label\High Mandatory Level            Label            S-1-16-12288

PS C:\Users\azureuser>
PS C:\Users\azureuser>
PS C:\Users\azureuser> arp -a

Interface: [redacted] --- 0x2
  Internet Address      Physical Address      Type
  [redacted]            12-34-56-78-9a-bc     dynamic
  [redacted]            ff-ff-ff-ff-ff-ff     static
  [redacted]            01-00-5e-00-00-16     static
  [redacted]            01-00-5e-00-00-fc     static
  [redacted]            ff-ff-ff-ff-ff-ff     static
PS C:\Users\azureuser>
PS C:\Users\azureuser>
PS C:\Users\azureuser> net user [redacted] PasswQrD2026 /add
The command completed successfully.

PS C:\Users\azureuser> net localgroup administrators [redacted] /add
The command completed successfully.

PS C:\Users\azureuser>
```

Innovations and Improvements

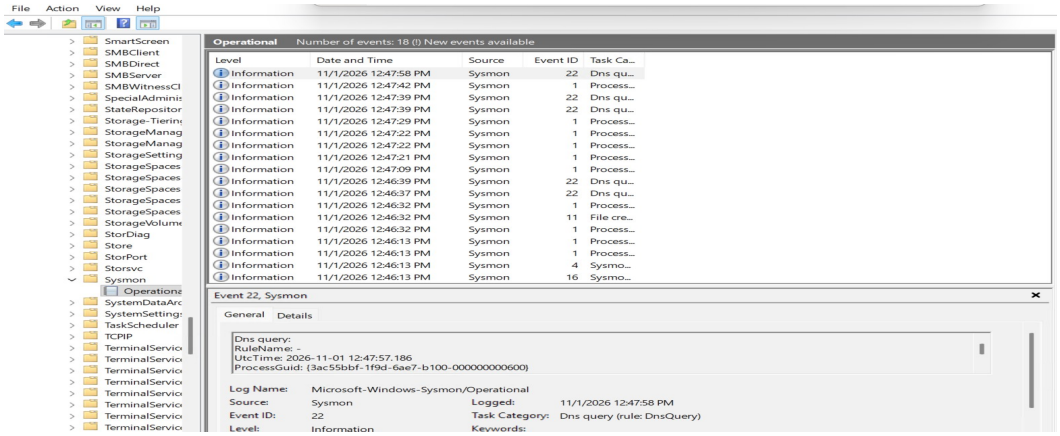
Sysmon Deployment

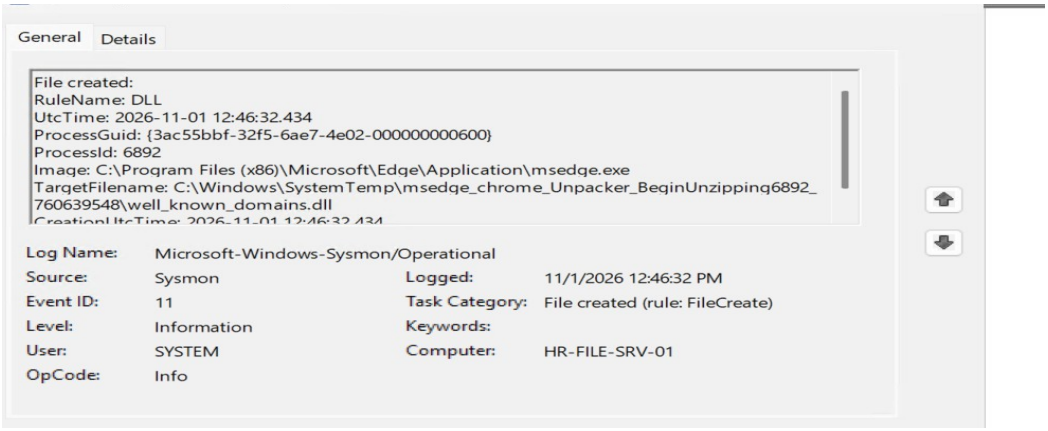
To improve forensic visibility, Sysmon was installed using the SwiftOnSecurity configuration.
Benefits:

- Process creation monitoring
- Network activity visibility
- File creation monitoring
- Improved threat hunting capabilities

Example Sysmon query:

```
SecurityEvent
| where EventSourceName == "Microsoft-Windows-Sysmon"
| where EventID == 1
| sort by TimeGenerated desc
```





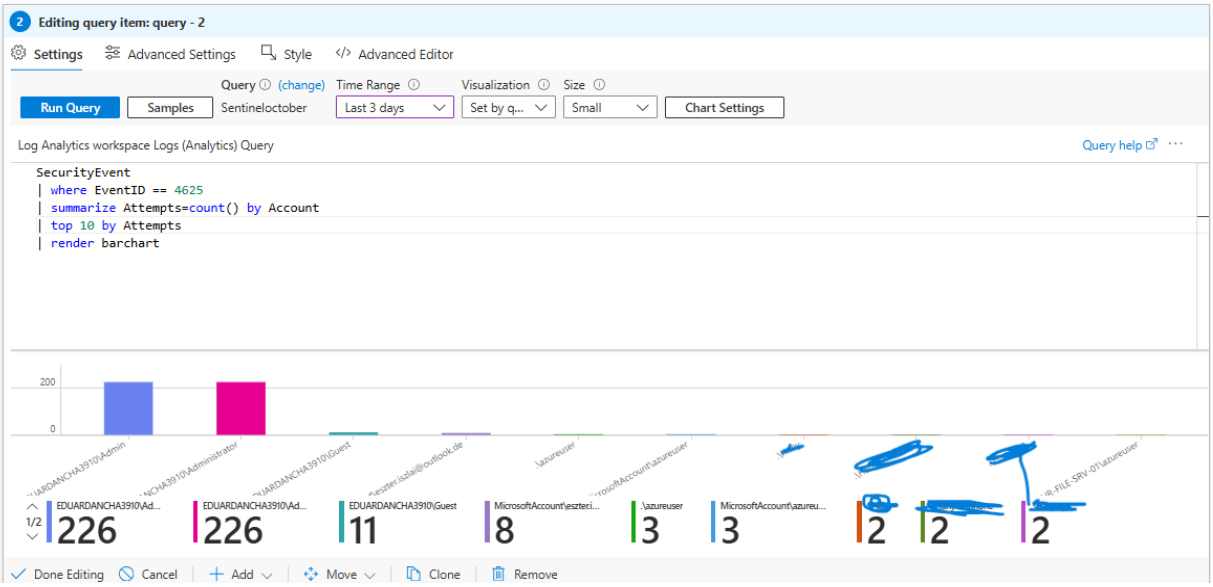
Sentinel Workbook Dashboard

A Sentinel Workbook dashboard was created to visualize activity.
The dashboard included:

- Successful logins
- Failed logins
- Top usernames
- Event statistics
- Sysmon activity

Example query:

```
SecurityEvent
| where EventID == 4625
| summarize Attempts=count() by Account
| top 10 by Attempts
```



Hardening Recommendations

The honeypot VM was intentionally exposed. To secure it in a production environment, the following actions are recommended:

Network Security

- Restrict RDP access using NSGs.
- Allow RDP only from trusted IP addresses.
- Use VPN access instead of public RDP.

Identity Security

- Enforce strong passwords.
- Enable Multi-Factor Authentication.
- Disable unused local accounts.

Monitoring

- Keep Sysmon installed.
- Continue forwarding logs to Sentinel.
- Create analytics rules for suspicious activities.

Least Privilege

- Remove unnecessary administrator rights.
- Monitor privileged accounts.

Threat Detection
Create Sentinel alerts for:

- Event ID 4720 (new account creation)
- Event ID 4732 (administrator group modifications)
- Excessive failed logins (4625)
- Unusual Sysmon process creation events

Conclusion

The project successfully demonstrated how Microsoft Sentinel can be used to detect, investigate, and analyze suspicious activity on a Windows honeypot. Through log analysis and KQL queries, it was possible to reconstruct a peer-to-peer attack, identify unauthorized account creation, and track privilege escalation.
The addition of Sysmon significantly improved telemetry collection and enhanced the ability to perform detailed forensic investigations.